Problem 8

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Problem 1 *Find the slope of the tangent line to the curve $y = f^{-1}(x)$ at $(4, 7)$ if the slope of the tangent line to the curve $y = f(x)$ at $(7, 4)$ is $\dfrac{2}{3}$.*

Explanation. Note that the statement "the slope of the tangent line to the curve $y = f(x)$ at $(7, 4)$ is $\dfrac{2}{3}$" specifically means that $f'(7) = \dfrac{2}{3}$. The slope of the tangent line to the curve $y = f^{-1}(x)$ at $(4, 7)$ is $(f^{-1})'(4)$, and so we use the formula for the derivative of the inverse function to compute: $(f^{-1})'(4) = \dfrac{1}{f'(7)} = \dfrac{1}{\frac{2}{3}} = \dfrac{3}{2}$.